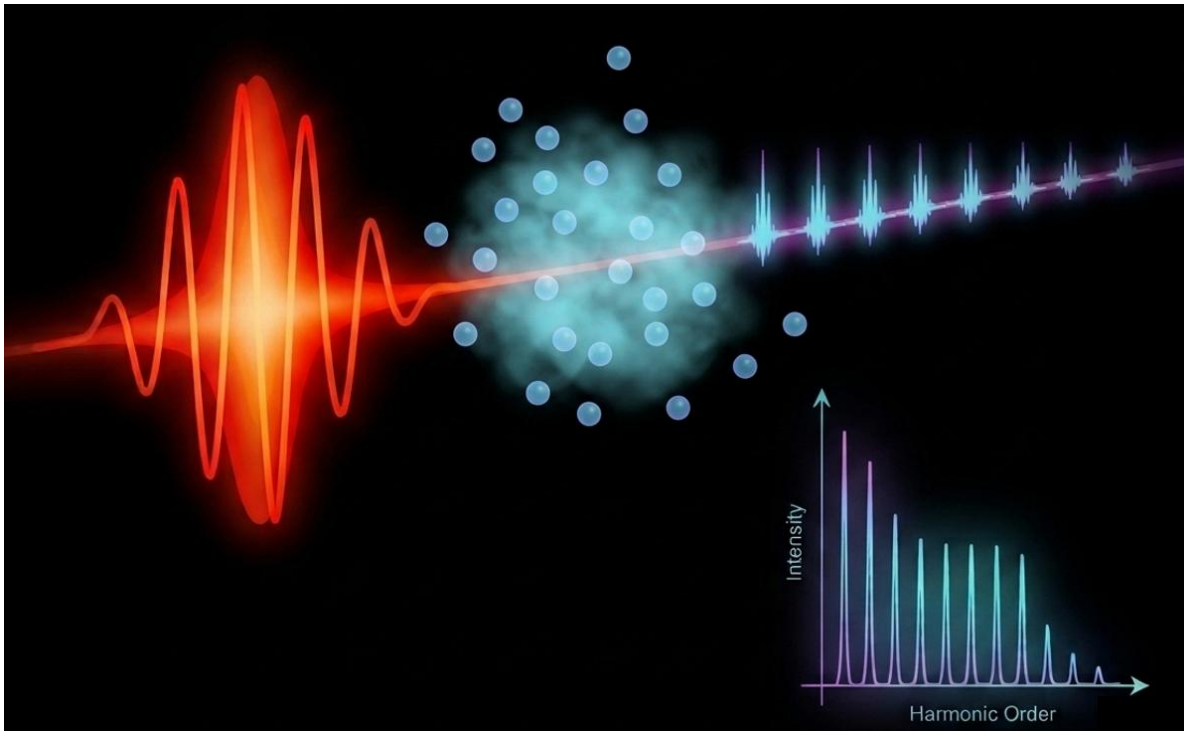


High harmonic generation and ultrafast electron dynamics simulations



IMPRS UFAST focus course

17th, 18th , 24th, 25th August
9:00-12:00h

Alba de las Heras
Theory department

UNTERSTÜTZT VON / SUPPORTED BY



Alexander von
HUMBOLDT
STIFTUNG



Outline

Session 1: Fundamentals of Strong-Field Physics

Session 2: Microscopic modeling of laser-driven electron dynamics and spectroscopic signatures in high harmonic generation

Session 3: Macroscopic picture of high harmonic generation and control of the properties of structured attosecond pulses

Session 4: Solid-state attosecond physics

Contents of Session 2:

Introduction to theory frameworks in strong-field physics

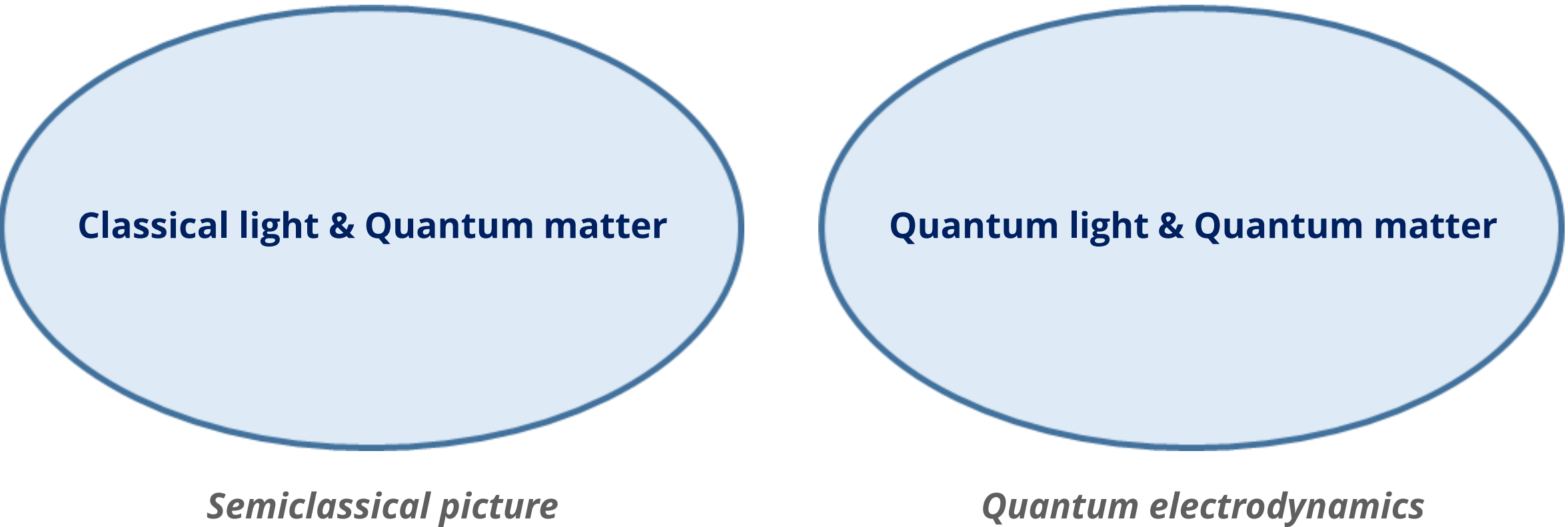
Time-Dependent Schrödinger Equation (TDSE) for the He atom

Two-electron correlated dynamics

Fundamentals of Time-Dependent Density-Functional Theory (TDDFT)

Octopus software

Introduction: Theory frameworks in laser-matter interactions



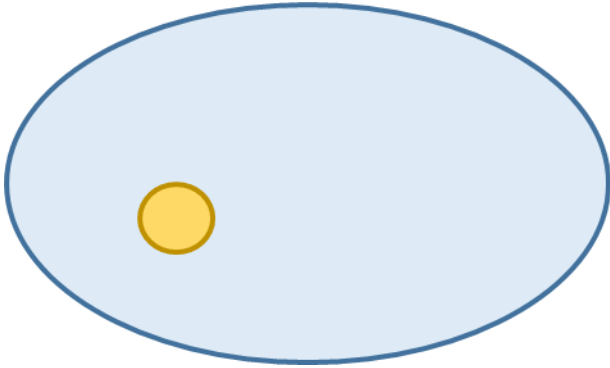
Classical light & Quantum matter

Semiclassical picture

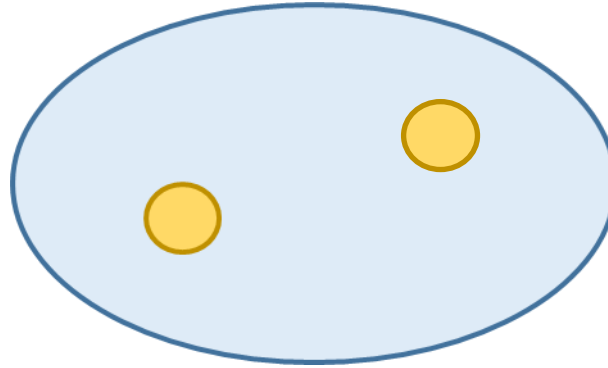
Quantum light & Quantum matter

Quantum electrodynamics

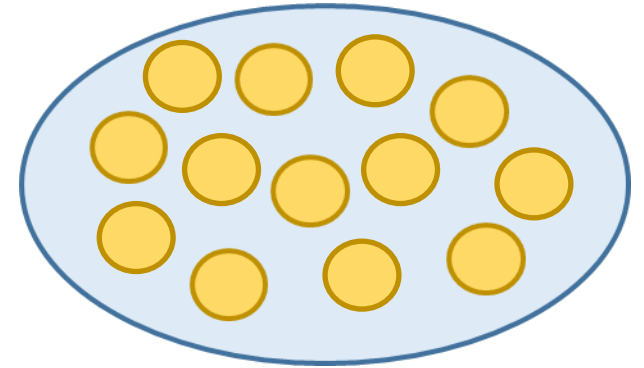
Introduction: Theory frameworks in laser-matter interactions



*Single Active-Electron
approximation*



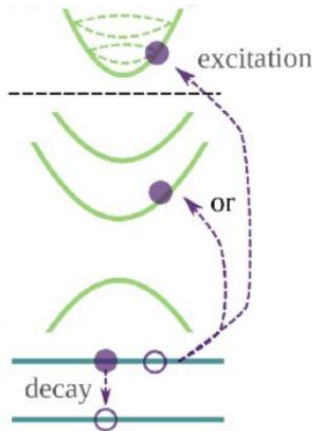
*Two Active-Electron
TDSE*



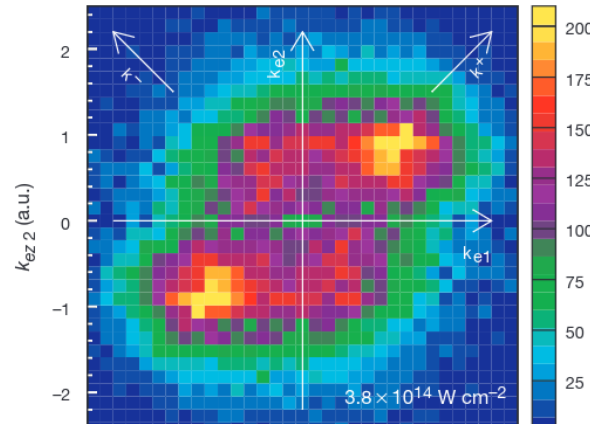
Multielectron methods:
TDDFT, TDHF, TDBSE...

Introduction: Examples of laser-driven multielectron dynamics

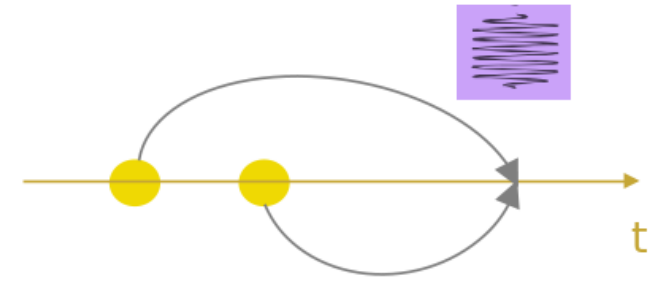
Electron rearrangements in photoexcitation



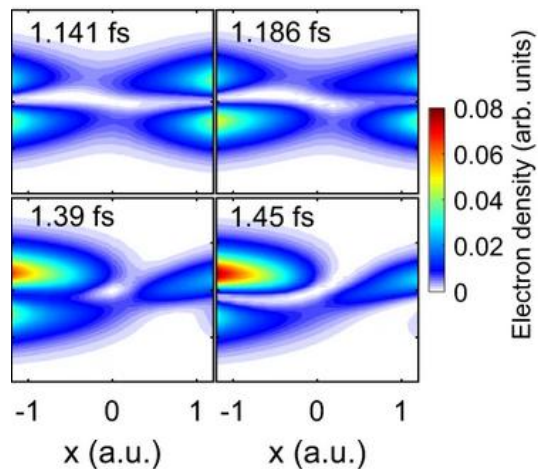
Multielectron ionization



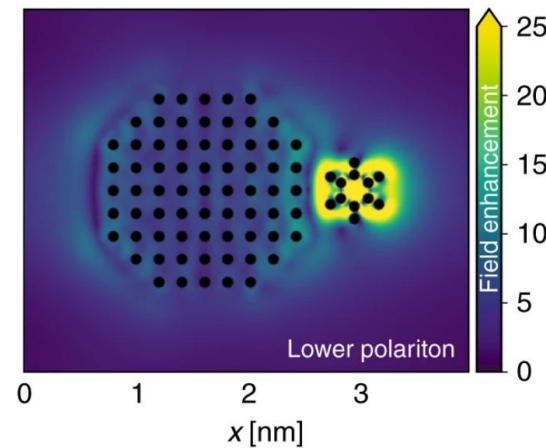
Multielectron recombination



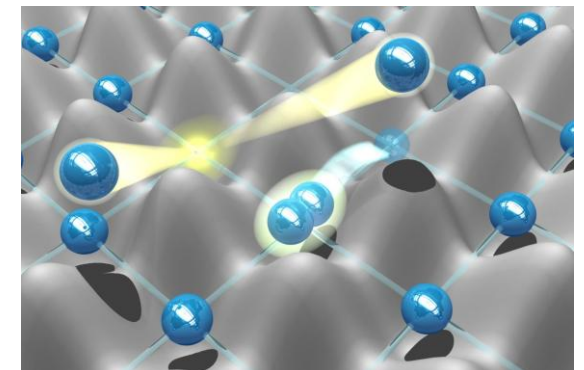
Charge migration



Collective electron dynamics



Effects from e-e correlation



Introduction: The many-body problem

Non-relativistic Hamiltonian of electrons in an atomic system interacting with a laser field:

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle \quad \text{TDSE}$$

$$\hat{H} = \sum_{j=1}^{N_e} \left\{ \frac{1}{2} \left(\hat{\mathbf{p}}_j + \frac{\mathbf{A}(\mathbf{r}_j, t)}{c} \right)^2 - \frac{Z_a}{|\mathbf{r}_j|} \right\} + \frac{1}{2} \sum_{j \neq l} \frac{1}{|\mathbf{r}_j - \mathbf{r}_l|}$$

Atomic units: $m_e = e = \hbar = a_0 = 1$

$$\mathbf{E}(\mathbf{r}_j, t) = -\frac{1}{c} \frac{\partial \mathbf{A}(\mathbf{r}_j, t)}{\partial t}$$

$$\mathbf{B}(\mathbf{r}_j, t) = \nabla \times \mathbf{A}(\mathbf{r}_j, t)$$

Introduction: The many-body problem

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Atomic units: $m_e = e = \hbar = a_0 = 1$

Dipole approximation:

Velocity gauge

$$\hat{H} = \sum_{j=1}^{N_e} \left\{ \frac{1}{2} \left(\hat{\mathbf{p}}_j + \frac{\mathbf{A}(t)}{c} \right)^2 - \frac{Z_a}{|\mathbf{r}_j|} \right\} + \frac{1}{2} \sum_{j \neq l} \frac{1}{|\mathbf{r}_j - \mathbf{r}_l|}$$

Length gauge

$$\hat{H} = \sum_{j=1}^{N_e} \left\{ \frac{1}{2} \hat{\mathbf{p}}_j^2 - \frac{Z_a}{|\mathbf{r}_j|} \right\} + \frac{1}{2} \sum_{j \neq l} \frac{1}{|\mathbf{r}_j - \mathbf{r}_l|} + \sum_{j=1}^{N_e} \mathbf{r}_j \cdot \mathbf{E}(t)$$

$$\mathbf{E}(\mathbf{r}_j, t) = -\frac{1}{c} \frac{\partial \mathbf{A}(\mathbf{r}_j, t)}{\partial t}$$

$$\mathbf{B}(\mathbf{r}_j, t) = \nabla \times \mathbf{A}(\mathbf{r}_j, t)$$

The dipole approximation is valid for non-relativistic electron energies and laser fields whose wavelength and spatial dependence are much larger than the atomic scale.

Introduction: Single-Active Electron approximation

The dynamics are described by an effective valence electron. Only one active electron interacts with the laser field whereas the other electrons are frozen in their initial state.

Time-dependent Schrödinger equation: 1D Hamiltonian of a single-active electron in an atomic system:

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle \quad \hat{H}^{SAE} = \frac{1}{2} \left(\hat{p}_x + \frac{A(t)}{c} \right)^2 - \frac{Z_a}{\sqrt{x^2 + a_{SAE}}},$$

Velocity gauge

✓ Significant simplification of the calculations

✗ Not appropriate for modeling multielectron dynamics

Full expression beyond the SAE approximation:

$$\hat{H} = \sum_{j=1}^{N_e} \left\{ \frac{1}{2} \left(\hat{\mathbf{p}}_j + \frac{\mathbf{A}(t)}{c} \right)^2 - \frac{Z_a}{|\mathbf{r}_j|} \right\} + \frac{1}{2} \sum_{j \neq l} \frac{1}{|\mathbf{r}_j - \mathbf{r}_l|} \quad \hat{H} = \sum_{j=1}^{N_e} \left\{ \frac{1}{2} \hat{\mathbf{p}}_j^2 - \frac{Z_a}{|\mathbf{r}_j|} \right\} + \frac{1}{2} \sum_{j \neq l} \frac{1}{|\mathbf{r}_j - \mathbf{r}_l|} + \sum_{j=1}^{N_e} \mathbf{r}_j \cdot \mathbf{E}(t)$$

Velocity gauge

Length gauge

Introduction: Strong-field approximation

Uses Green's propagator formalism:

$$|\Psi(t)\rangle = i\hat{G}(t, t_0) |\Psi(t_0)\rangle = e^{-i \int_{t_0}^t \hat{H}(t') dt'} |\Psi(t_0)\rangle$$

Key points:

- Typically constructed as a single-electron approach
- Neglects the binding potential once the electron is ionized
- It only considers the coupling of a single bound electron state with the continuum, discarding excitation to other states.

$$\begin{aligned} |\Psi^{SFA}(t)\rangle &= |\varphi_0(t)\rangle + |\delta\Psi(t)\rangle \\ &= i\hat{G}_A(t, t_0)|\varphi_0\rangle + i \int_{t_0}^t \hat{G}_F(t, t') \hat{H}_{int}(t') \hat{G}_A(t', t_0) |\varphi_0\rangle dt' \end{aligned}$$

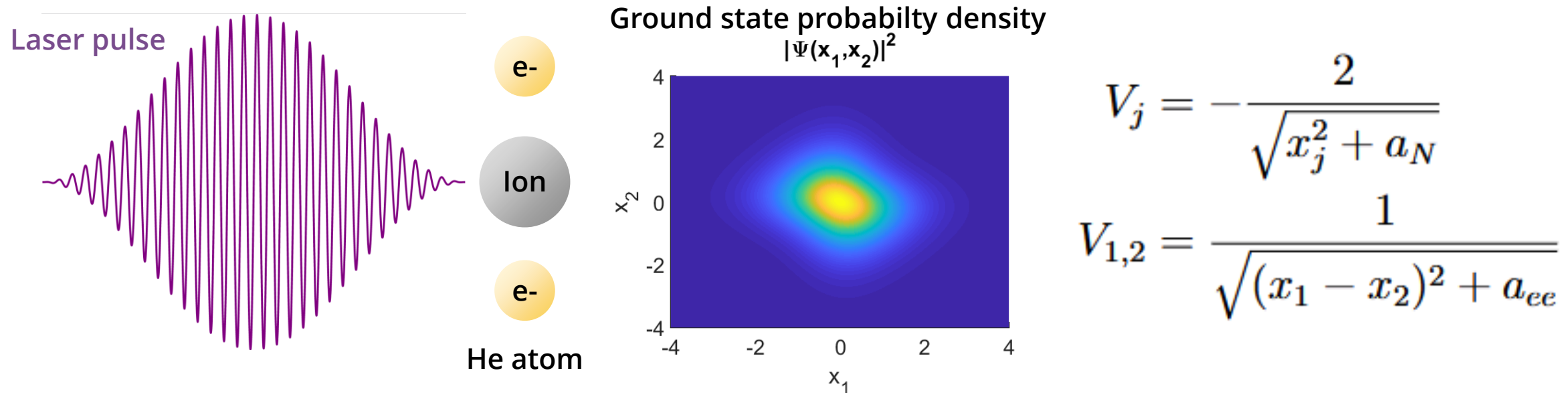
$$|\varphi_0(t)\rangle = i\hat{G}_A(t, t_0)|\varphi_0\rangle = e^{-i\epsilon_0(t-t_0)} |\varphi_0\rangle$$

$$|\delta\Psi(t)\rangle = -i \int \int_{t_0}^t e^{-i\mathcal{K}(\mathbf{p}_{kin}, t, t')} e^{-i\epsilon_0(t'-t_0)} \langle \varphi_{\hat{\mathbf{p}}} | \hat{H}_{int}(t') | \varphi_0 \rangle dt' |\varphi_{\hat{\mathbf{p}}}\rangle d^3p$$

1D Time-Dependent Schrödinger Equation for the He atom

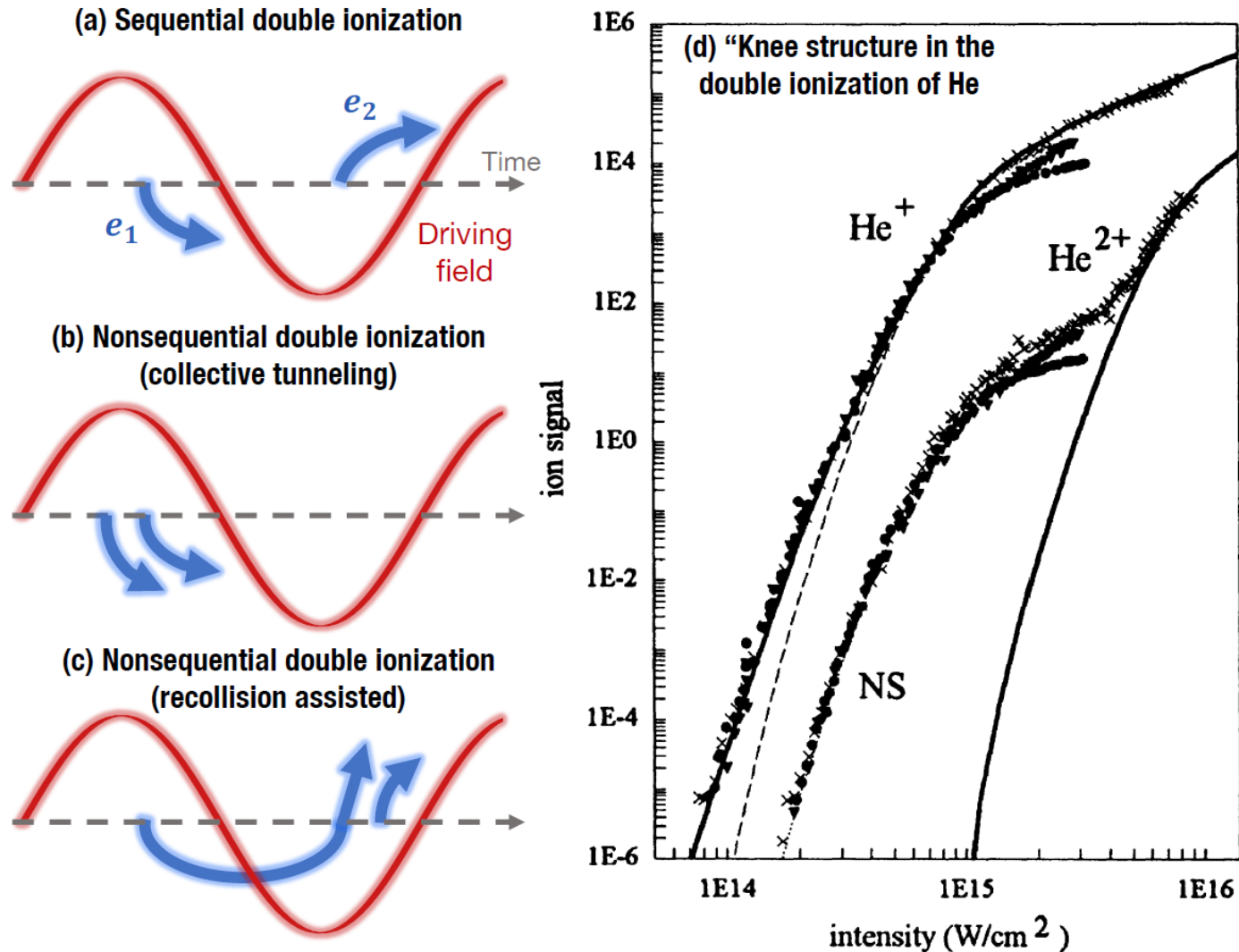
Ab initio formalism: we consider the two active electrons interacting with the laser field:

$$i\frac{\partial}{\partial t}\Psi(x_1, x_2, t) = \left[\sum_{j=1}^2 \left\{ \frac{1}{2} \left(\hat{p}_{x,j} + \frac{A(t)}{c} \right)^2 + V_j \right\} + V_{1,2} \right] \Psi(x_1, x_2, t)$$

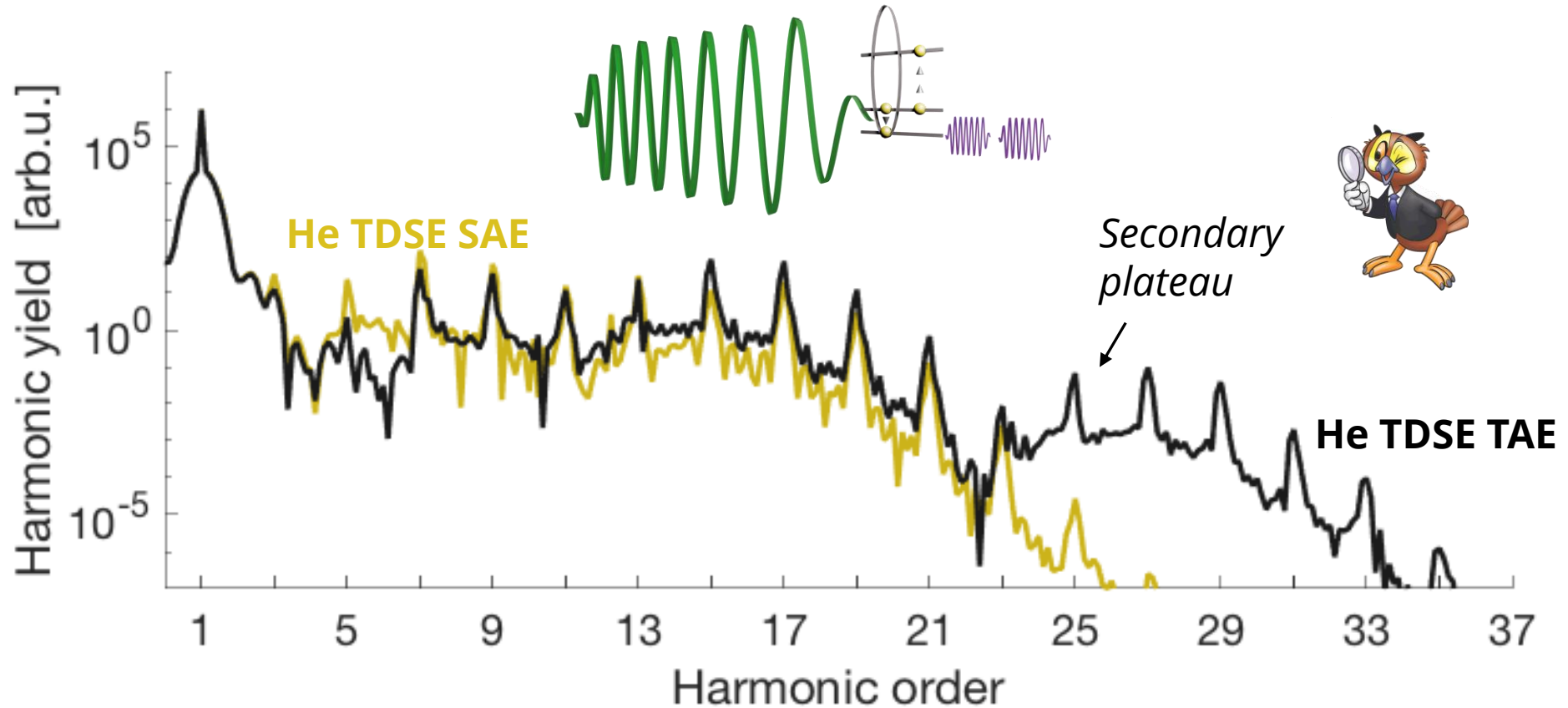


We model the dynamics along the direction of a linearly polarized laser field

Two-electron correlated dynamics: double ionization



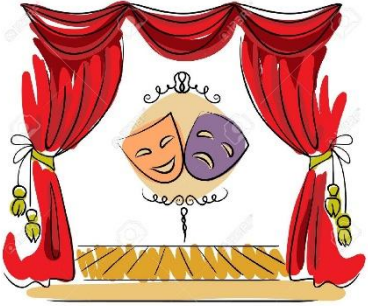
Two-electron correlated dynamics: correlated back-reaction



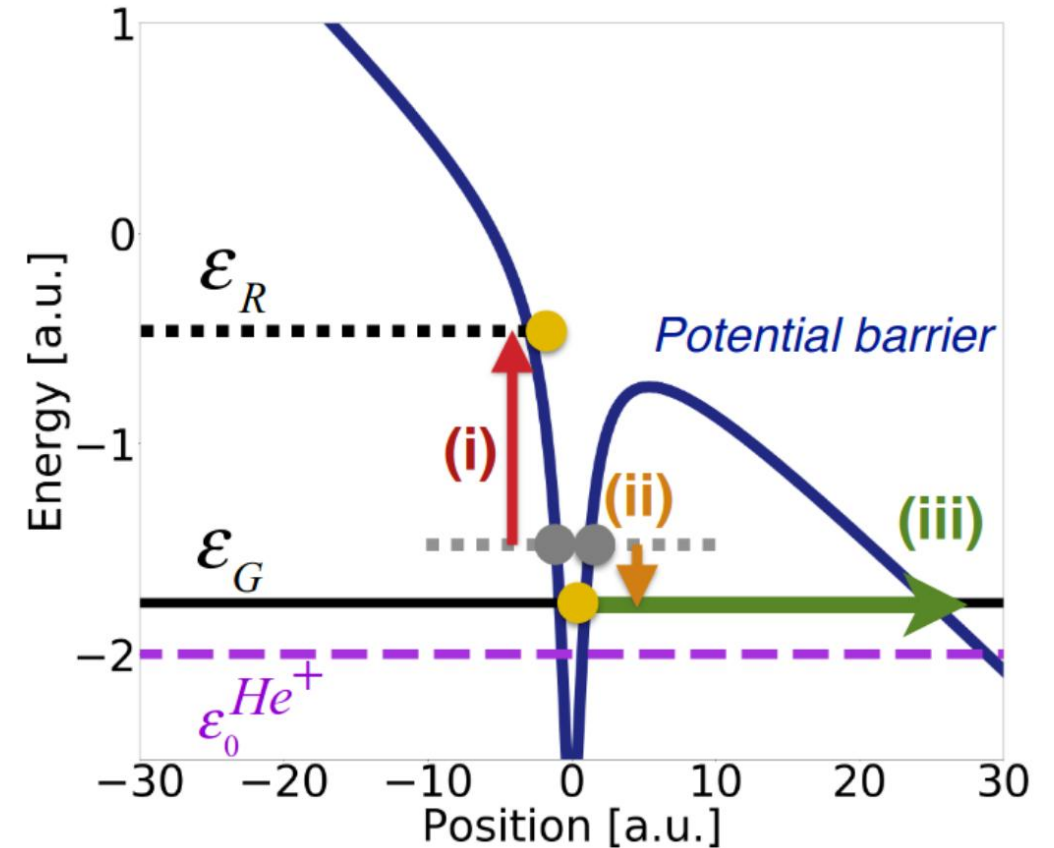
A secondary plateau in high harmonic generation from He emerges only in the TAE calculation

A. de las Heras *et al.* *Phys. Rev. Research* (2020)

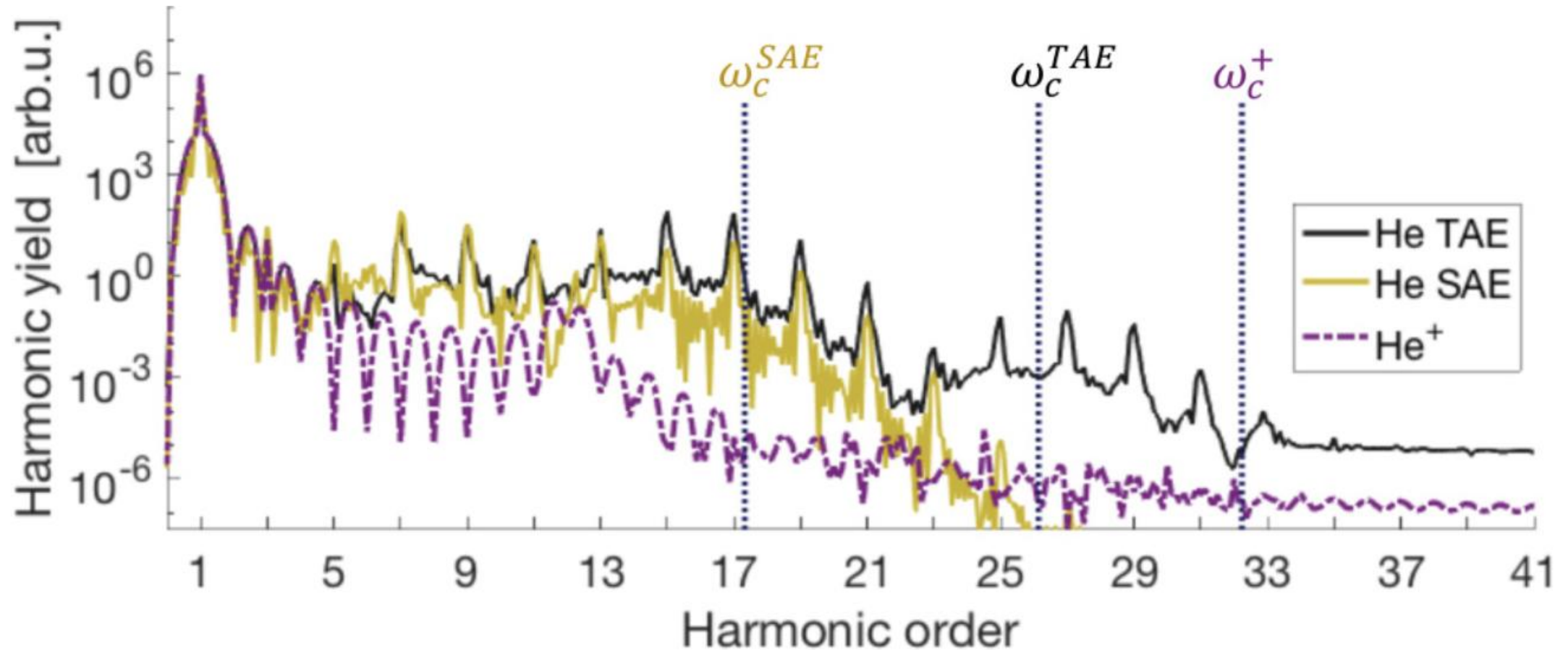
The correlated back reaction in HHG



1. The primary electron absorbs the energy from the laser field and induces the excitation of the secondary electron.
2. **Correlated back-reaction**: knock-down of the primary electron to a deeper energy state.
3. The primary electron starts the usual process of high harmonic generation from a deeper energy state.



The correlated back reaction in HHG

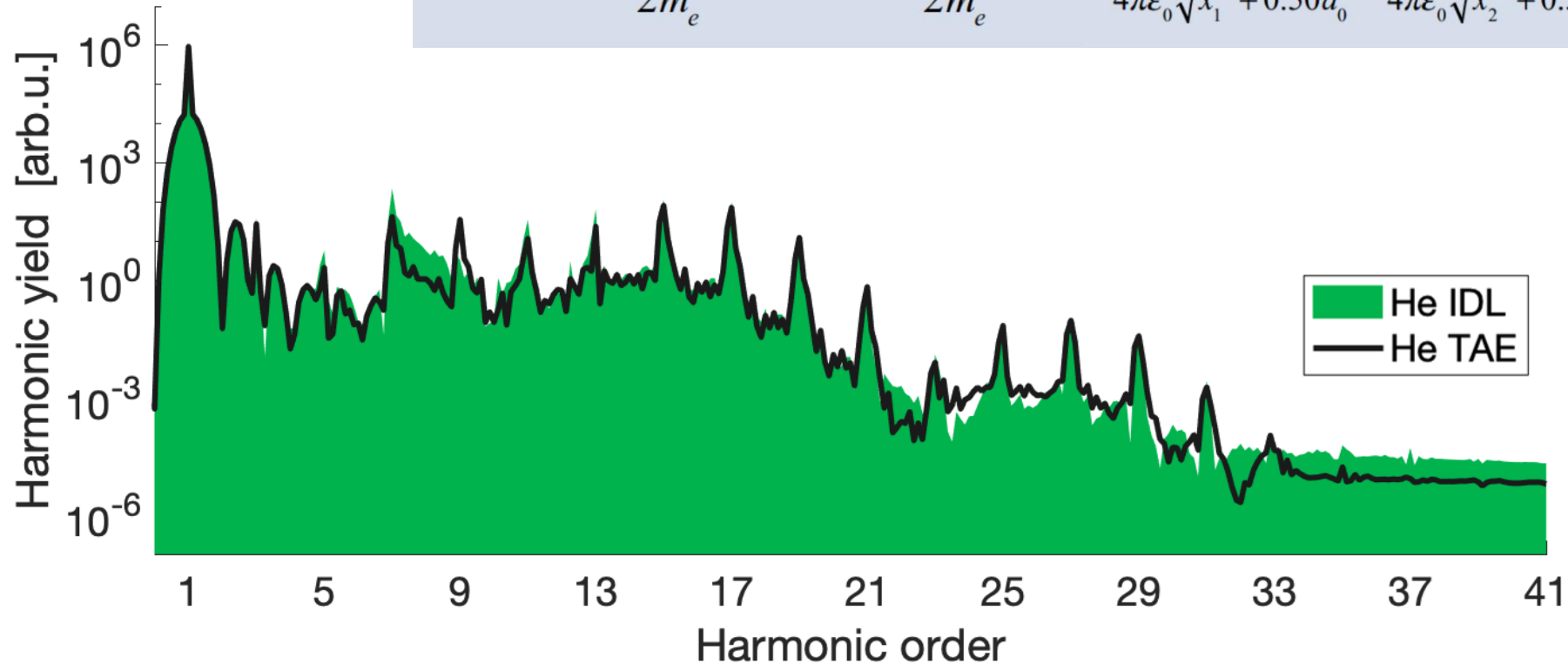


Idle electron model

He IDL: Idle electron (Two correlated electrons but only one interacts with the laser)



$$\hat{H}^{IDL} = \frac{\left(\hat{p}_1 + e A(t)\right)^2}{2m_e} + \frac{\left(\hat{p}_2 + e A \text{ (crossed out)}\right)^2}{2m_e} - \frac{2e^2}{4\pi\epsilon_0\sqrt{x_1^2 + 0.50a_0}} - \frac{2e^2}{4\pi\epsilon_0\sqrt{x_2^2 + 0.50a_0}} + \frac{e^2}{4\pi\epsilon_0\sqrt{(x_1 - x_2)^2 + 0.32a_0}}$$



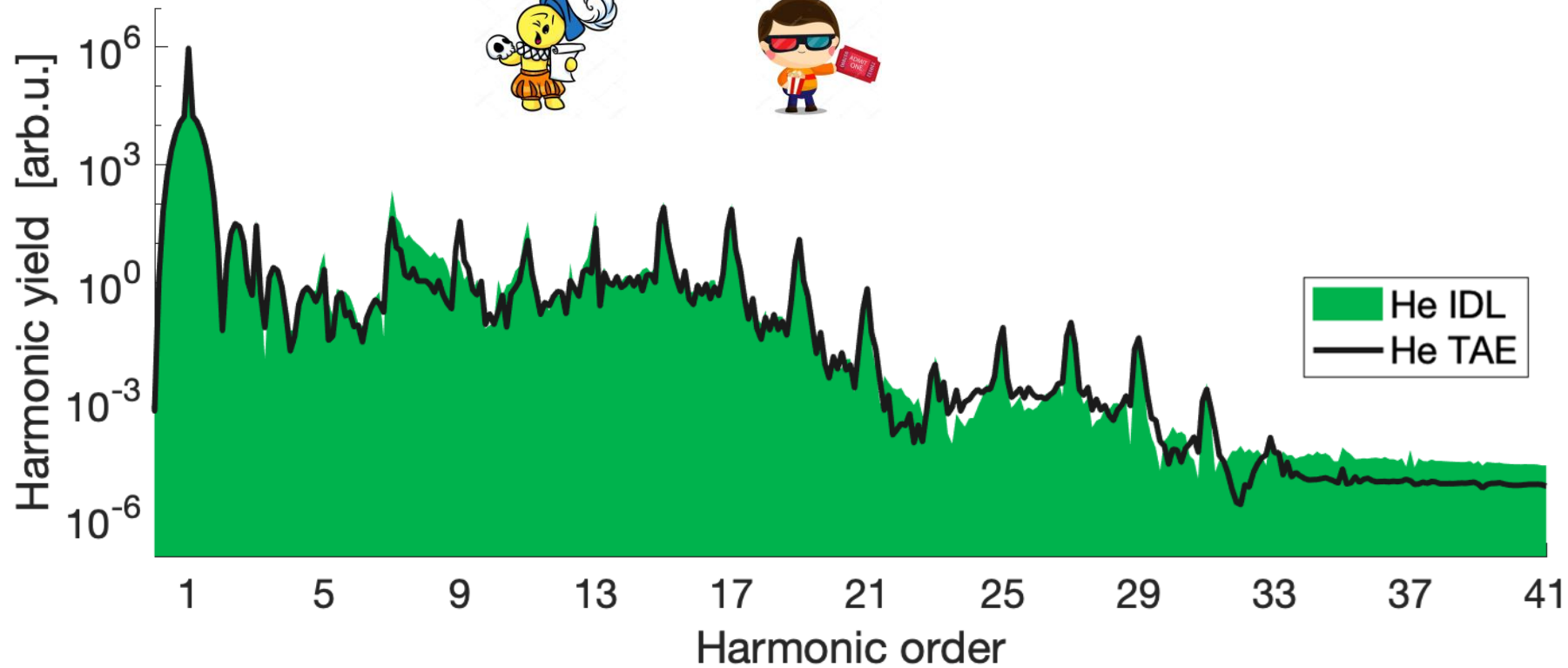
Idle electron model

Electrons with “active” and “passive” roles

*Primary
electron*



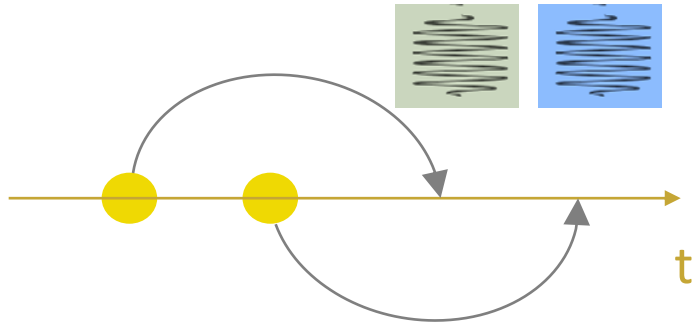
*Secondary
electron*



A. de las Heras *et al.* *Phys. Rev. Research* (2020)

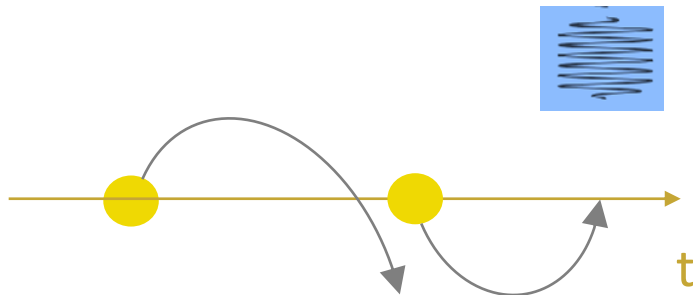
Two-electron mechanisms can extend the energy cutoff in HHG

Sequential double recombination



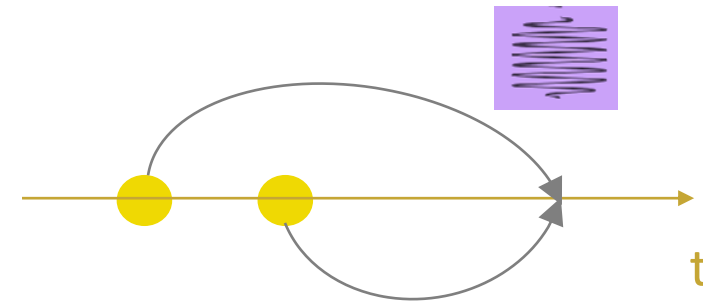
D. Lappas & R. van Leeuwen.
Journal of Physics B (1996)

Double ionization & single recombination



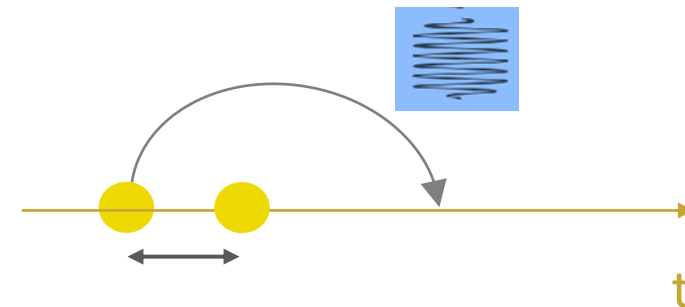
I. Tikhomirov, T. Sato & K. L. Ishikawa.
Phys. Rev. Lett. (2017)

Nonsequential double recombination



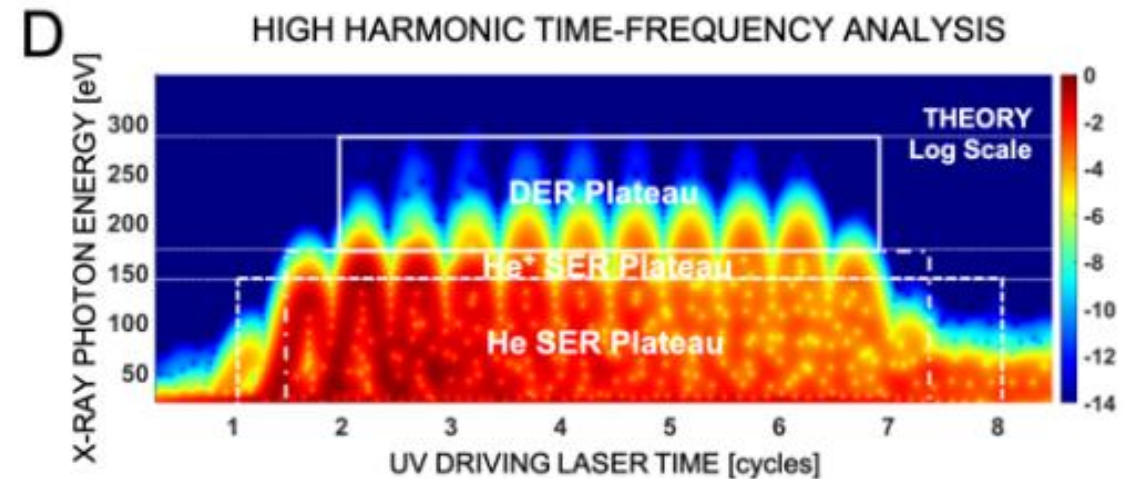
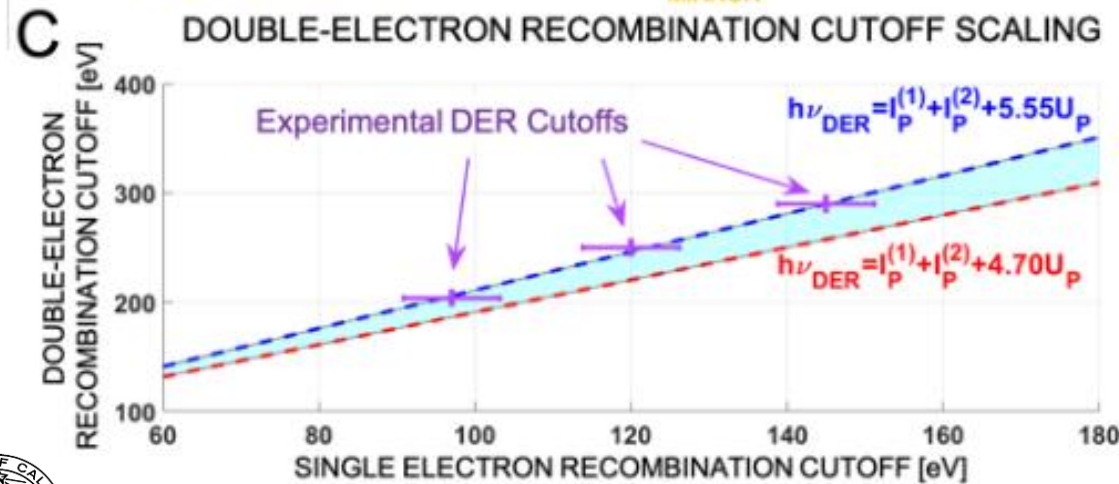
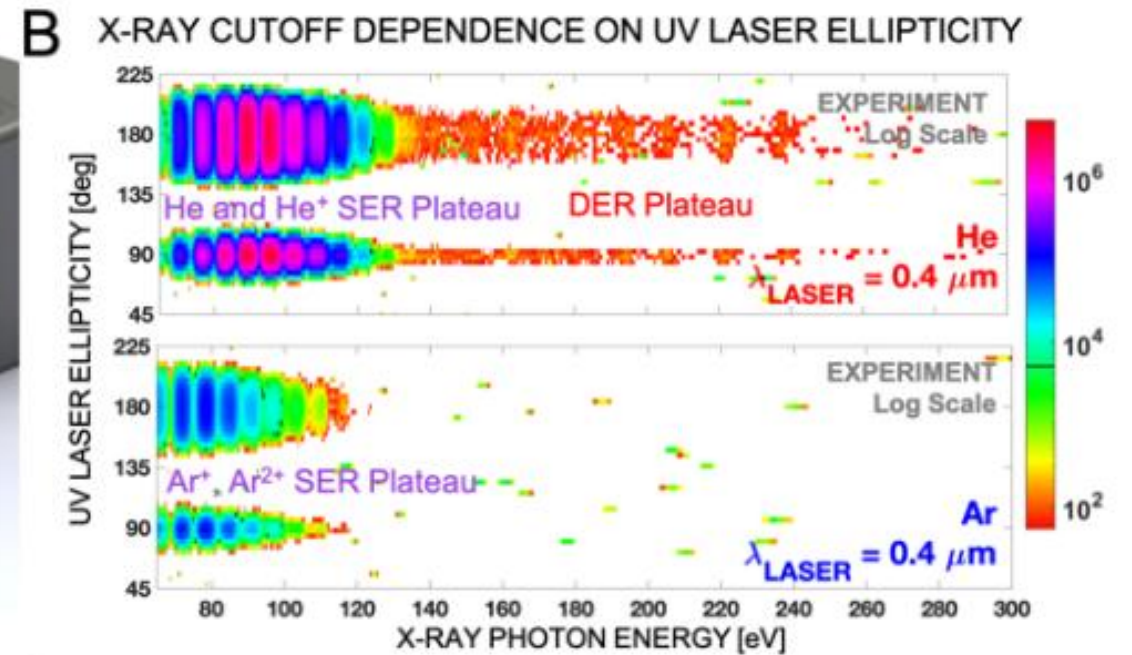
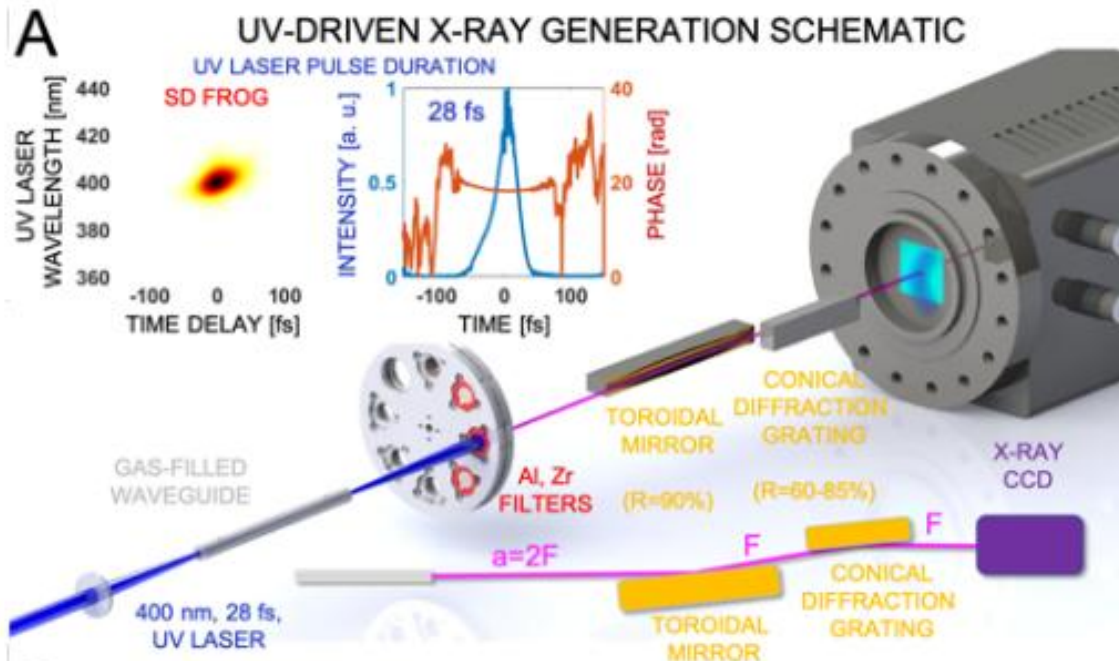
P. Koval et al. *Phys. Rev. Lett.* (2007)
S. Wang, J. Yan, A. de las Heras et al. *Nat. Photonics* 2026

Single ionization & single recombination



A. de las Heras, C. Hernández-García, & L. Plaja,
Phys. Rev. Res. (2020)

Two-electron correlated dynamics: double electron recombination



Fundamentals of TDDFT (Time-Dependent Density-Functional Theory)

- Every observable of a quantum system can be calculated from the density. The electron density is defined as the probability of finding an electron of spin s_1 at a position $\mathbf{r}_1$ and time t :

$$n(\mathbf{r}, t) = N_e \sum_{s_1, s_2, \dots, s_{N_e}} \int \dots \int |\psi(\mathbf{r}s_1, \mathbf{r}_2s_2, \dots, \mathbf{r}_{N_e}s_{N_e}, t)|^2 d^3r_2 \dots d^3r_{N_e}.$$

- The electron density of the many-electron interacting system can be obtained by solving a fictitious problem of independent electrons in an effective potential.
- Time-dependent Kohn-Sham equations (TDKS):

$$i\frac{\partial}{\partial t}|\varphi_j^{KS}(t)\rangle = \left[\frac{1}{2} \left(-i\nabla + \frac{\mathbf{A}(t)}{c} \right)^2 + V_{KS}[n](\mathbf{r}, t) \right] |\varphi_j^{KS}(t)\rangle$$

Hartree potential

$$V_{KS}[n](\mathbf{r}, t) = V_{ion}(\mathbf{r}) + V_H[n](\mathbf{r}, t) + \underbrace{V_{xc}[n](\mathbf{r}, t)}_{\text{Unknown!!}}$$

$$V_H[n](\mathbf{r}, t) = \int d^3r' \frac{n(\mathbf{r}', t)}{|\mathbf{r} - \mathbf{r}'|}$$

$$n(\mathbf{r}, t) = \sum_j^{N_e} |\langle \mathbf{r} | \varphi_j^{KS}(t) \rangle|^2 = \sum_j^{N_e} |\varphi_j^{KS}(\mathbf{r}, t)|^2$$

Average Coulomb repulsion between electron pairs

Fundamentals of TDDFT (Time-Dependent Density-Functional Theory)

- The explicit expression of $V_{xc}[n](\mathbf{r}, t)$ remains unknown, and thus approximations are unavoidable. Still, two different contributions can be distinguished:

$$V_{xc}[n](\mathbf{r}, t) = V_x[n](\mathbf{r}, t) + V_c[n](\mathbf{r}, t).$$

- The exchange contribution, $V_x[n](\mathbf{r}, t)$, arises due to the antisymmetry of the electron wavefunction dictated by the Pauli exclusion principle (establishing that two identical fermions cannot occupy simultaneously the same quantum state). The effect of the exchange potential is a repulsion between electrons with the same spin. It also removes the self-interaction of an electron with itself, which has no physical sense.
- The correlation potential, $V_c[n](\mathbf{r}, t)$, accounts for the remaining complex multi-electron interactions beyond the simple repulsion due to their charge or spin, such as electron localization, screening effects, or non-local behaviors.

Fundamentals of TDDFT (Time-Dependent Density-Functional Theory)

- **The Adiabatic Local Density Approximation (ALDA)** is the standard approximation for the exchange-correlation (XC) functional in TDDFT.

- In the adiabatic approximation, the instantaneous density is the input in a static XC Functional:

$$V_{XC}[n](\mathbf{r}, t) \approx V_{XC}^{ground}[n(\mathbf{r}, t)]$$

It assumes that the XC potential depends only on the instantaneous density and not on its past history.

- The local density approximation assumes that the XC potential at a given point in space depends only on the local electron density at that point. As a benchmark model to calculate the XC energy, a homogeneous electron gas is considered.

Fundamentals of TDDFT (Time-Dependent Density-Functional Theory)

- **The ground state** is obtained from self-consistent calculations using the (time-independent) Kohn Sham equations:

$$\left(-\frac{1}{2}\nabla^2 + V_{KS}^0[n_0](\mathbf{r}) \right) \varphi_j^0(\mathbf{r}) = E_j \varphi_j^0(\mathbf{r})$$

$$n_0(\mathbf{r}) = \sum_{j=1}^{N_e} |\varphi_j^0(\mathbf{r})|^2$$

$$V_{KS}^0[n_0](\mathbf{r}) = V_{ion}(\mathbf{r}) + V_H^0[n_0](\mathbf{r}) + V_{xc}^0[n_0](\mathbf{r})$$

Octopus software



All the info in <https://octopus-code.org>

- Octopus is an **open-access** scientific software designed for **first-principles simulations of quantum systems**. It allows the computation of the electronic structure or dynamics using DFT or TDDFT. The Kohn Sham equations are solved in a real space grid for each time iteration.
- For time-dependent problems, we prepare 2 input files: one for the ground state calculation and another one to propagate it in time. We will name them “inp_gs” and “inp_td” respectively.
- Parameters such as the spatial and time steps, size of the numerical box, etc. need to be converged.
- For calculations of the band structure of a solid, we prepare 2 input files (“inp_gs” and “inp_bands”). In this case, we also tune the k-points grid for convergence.
- In the input files, we indicate the output quantities and choose the calculation options for a certain system.
- Some calculations require computer clusters, but simpler ones can run on a personal computer.

Exercises with Octopus software



E1. Generate the Octopus input files to calculate the temporal evolution of the **total current** of **Neon** atom driven by a circularly polarized laser pulse of 4 cycles full duration, 267 nm central wavelength and $1.4 \times 10^{15} \text{ W/cm}^2$ peak intensity. The spatial mesh should be chosen with a step of 0.4 a.u. in a cylindrical box shape of radius 25 a.u. and length 25 a.u. For the time propagation, set a time step of 0.02 a.u. and add an absorbing boundary extending 20 a.u. to avoid artificial reflections at the ends of the numerical box. Output the current density in the plane of the electric field polarization at 5 instants of time.

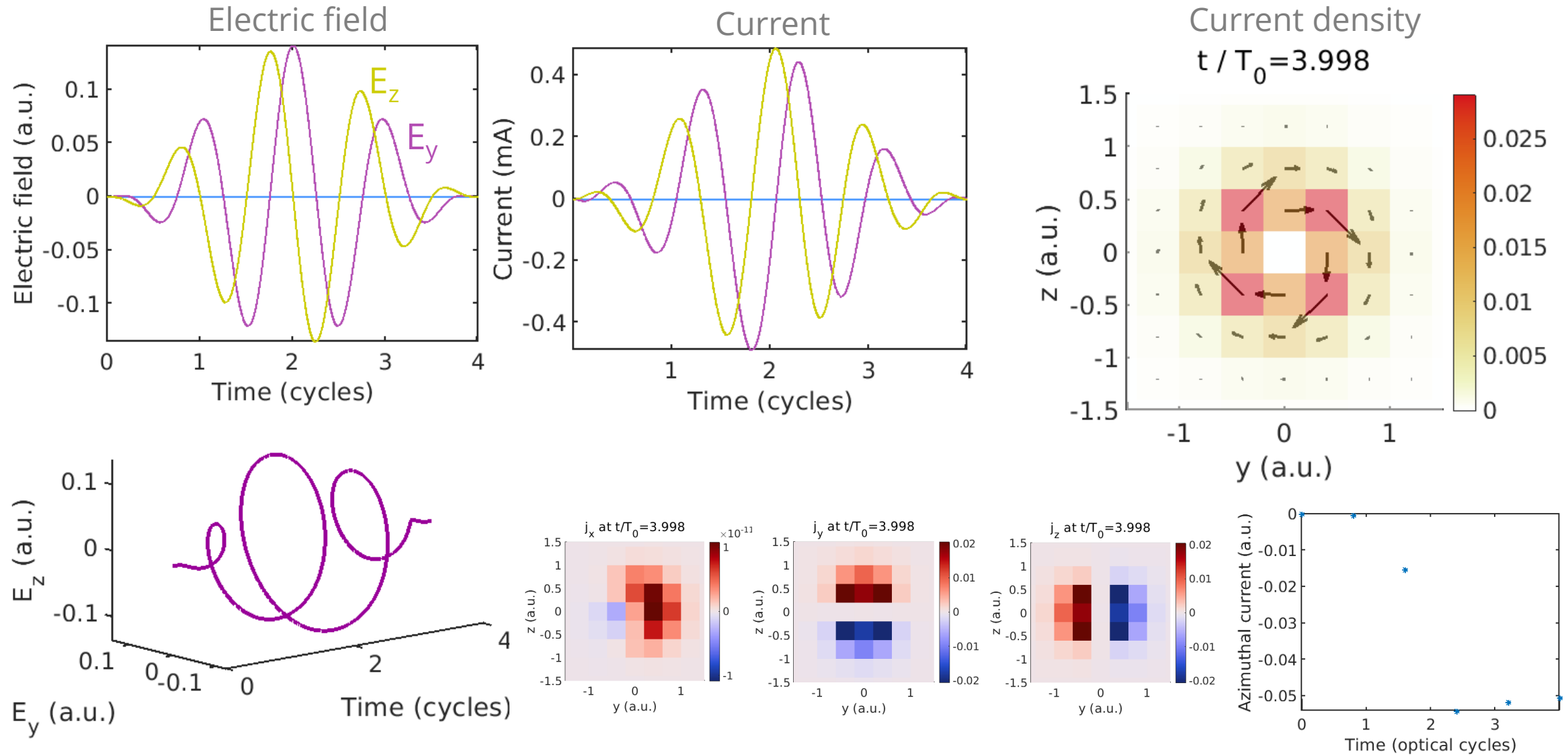
E2. Generate the Octopus input files to calculate the **high harmonic spectrum** of **Neon** atom driven by a linearly polarized laser pulse of 6 cycles full duration, 400 nm central wavelength and $2 \times 10^{14} \text{ W/cm}^2$ peak intensity. The spatial mesh should be chosen with a step of 0.4 a.u. in a cylindrical box shape of radius 25 a.u. and length 25 a.u. For the time propagation, set a time step of $dt=0.02$ a.u. and add an absorbing boundary extending 20 a.u. to avoid artificial reflections in the numerical box. (Optional: Freeze the XC potential for faster computation).

Important files after executions with Octopus software



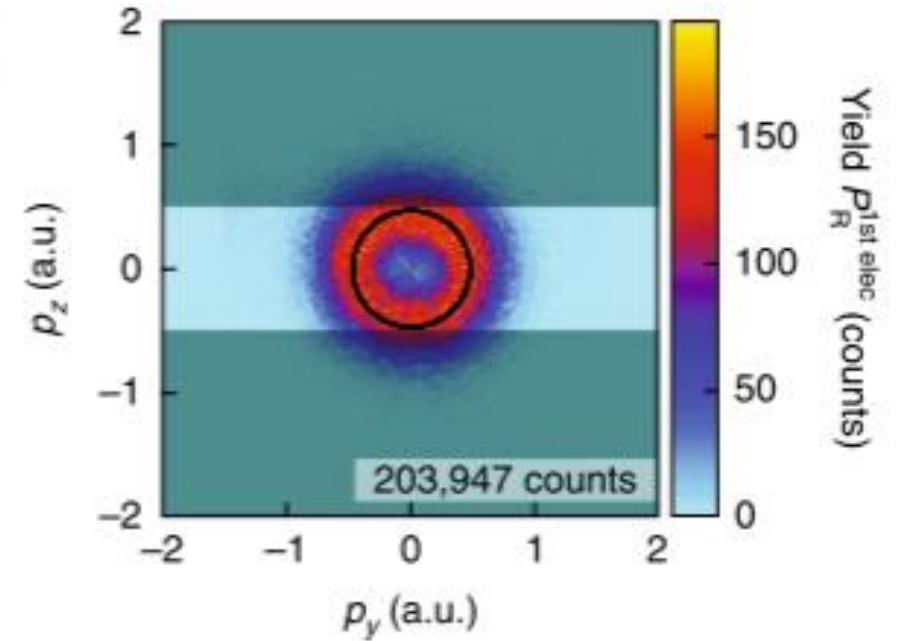
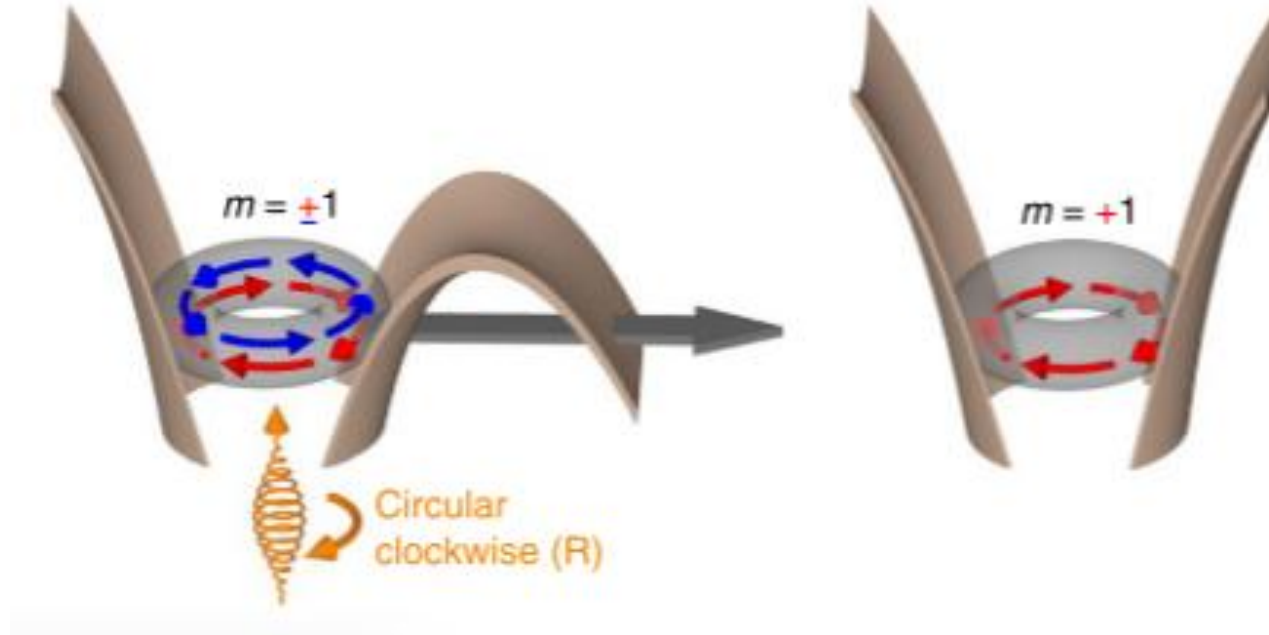
- General information about the execution in log file
- All the input variables are in `exec/parser.log`
- In ground state calculations, you can find information about the convergence in `static/convergence` and all the calculated ground state parameters in `static/info`. In solids, the crystal structure (geometry output) is stored in `static/geometry` and `static/crystal`
- After running an “unocc” calculation to compute the band structure, you will get the data in `static/bandstructure`
- In time-dependent calculations, the folder `td.general` contains the “TDOutput” data and `output_iter` the “Output” data at the specified instants of time

E1: Currents in Ne atom driven by a CP driving pulse



Long-lived ring-current from laser-atom interaction

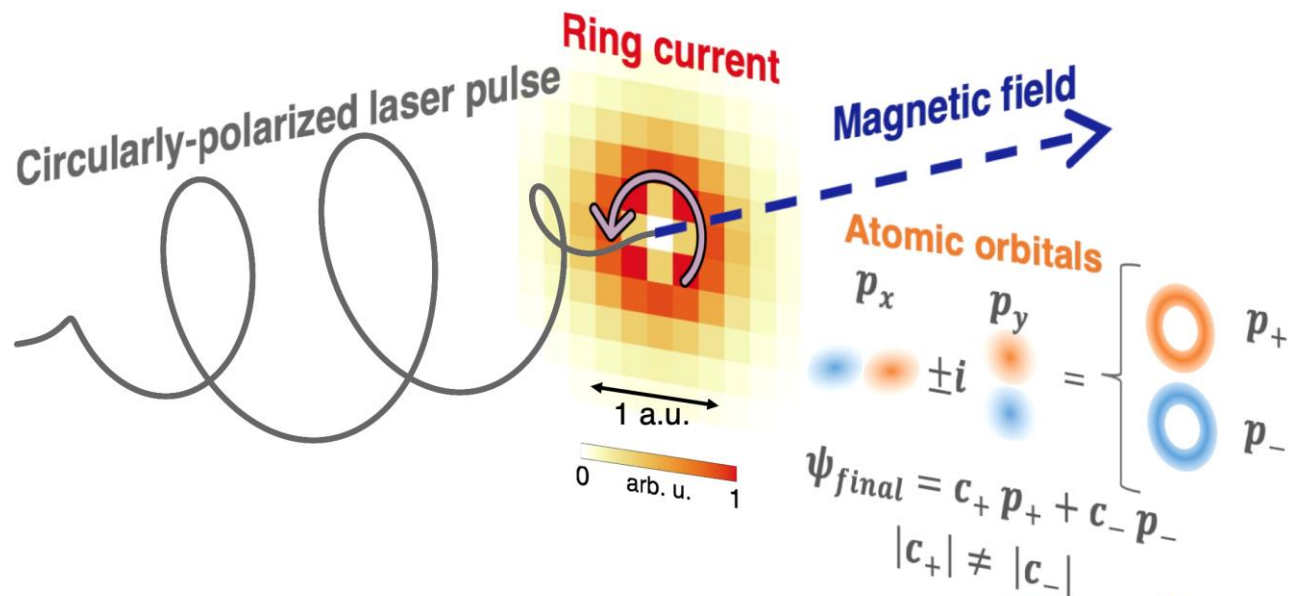
Experimental ring-current measurements on Ar atoms



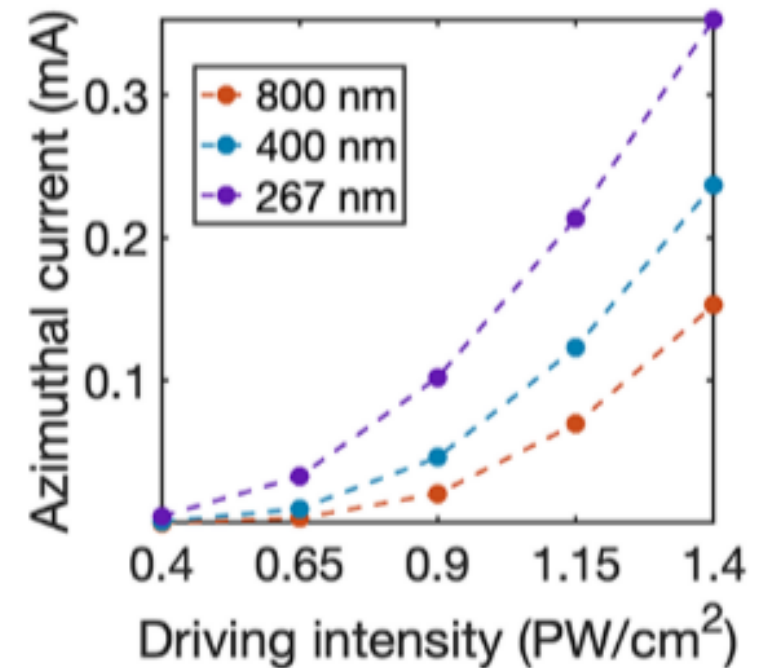
S. Eckart, et al. *Nat. Physics* 14, 701 (2018)

Theoretical fundamentals: I. Barth, O. Smirnova. *Phys. Rev. A* 84, 063415 (2011)

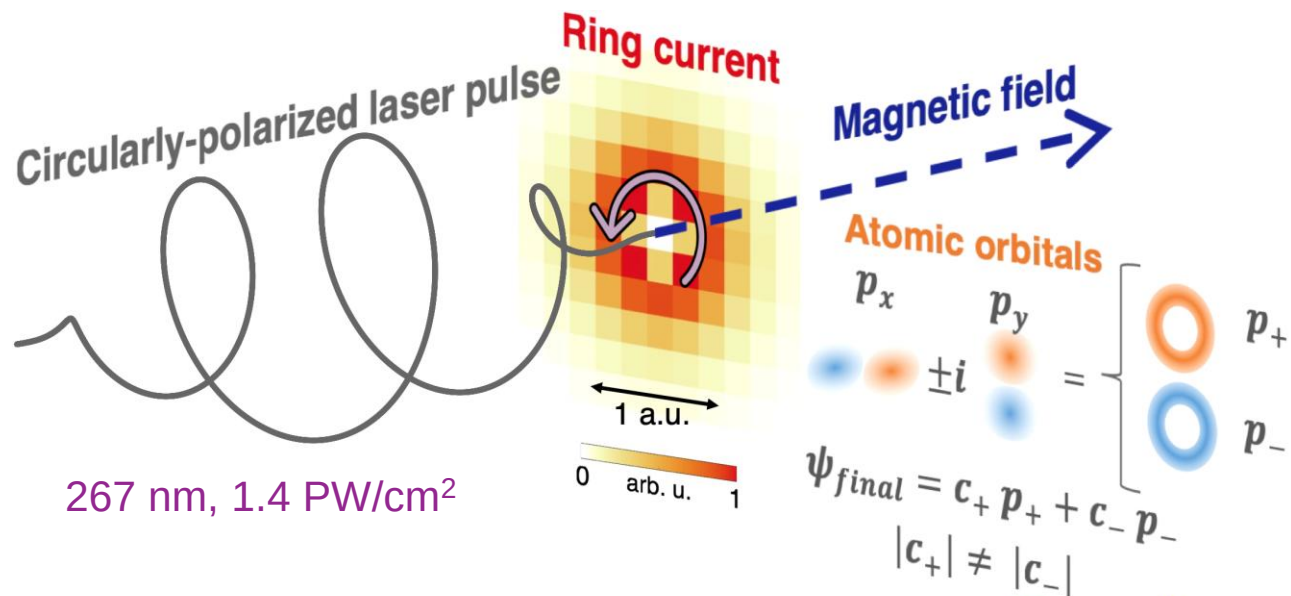
Long-lived ring-current from laser-atom interaction



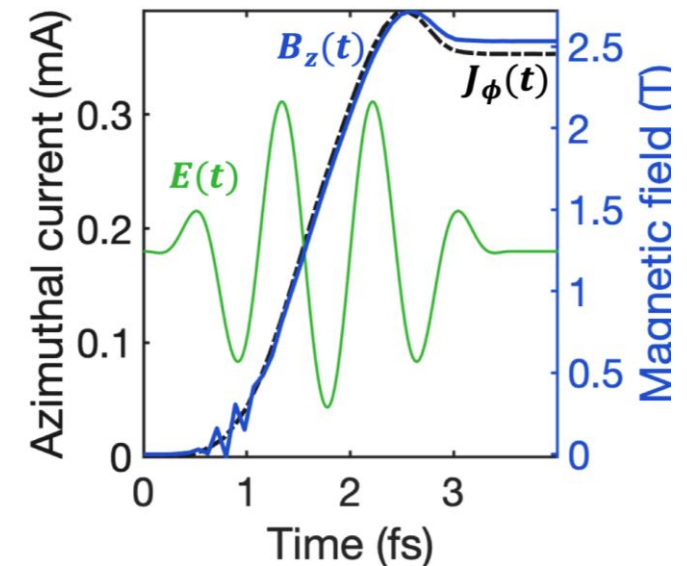
Dependence of the current on the driving intensity and wavelength



Long-lived magnetic field from laser-atom interaction



Strong long-lived magnetic field



Ab-initio numerical methods: TDDFT+Maxwell solvers in Octopus

Computation of electron dynamics with **time-dependent density-functional theory**

Time-dependent Kohn Sham equations in the dipole approximation

$$i\frac{\partial}{\partial t}|\varphi_j^{KS}(t)\rangle = \left[-\frac{1}{2}\nabla^2 + V_{KS}(\mathbf{r}, t) + \mathbf{r} \cdot \mathbf{E}(t)\right] |\varphi_j^{KS}(t)\rangle$$

$\mathbf{E}(t)$ is the electric field of the laser pulse

$$\mathbf{E}(t) = -\frac{1}{c} \frac{\partial \mathbf{A}(t)}{\partial t}$$

$$V_{KS}(\mathbf{r}, t) = V_{ion}(\mathbf{r}) + V_H(\mathbf{r}, t) + V_{xc}[n](\mathbf{r}, t)$$

Exchange-correlation functional in the adiabatic local density approximation (ALDA)

Current density

$$\mathbf{j}(\mathbf{r}, t) = \frac{1}{2} \varphi_n^{KS,*}(\mathbf{r}, t) \left(-i\nabla + \frac{\mathbf{A}(t)}{c} - i[v_{ion}, r] \right) \varphi_n^{KS}(\mathbf{r}, t)$$

Computation of the generated electromagnetic fields with **Maxwell's equations**

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

$$\nabla \cdot \mathbf{B} = 0$$

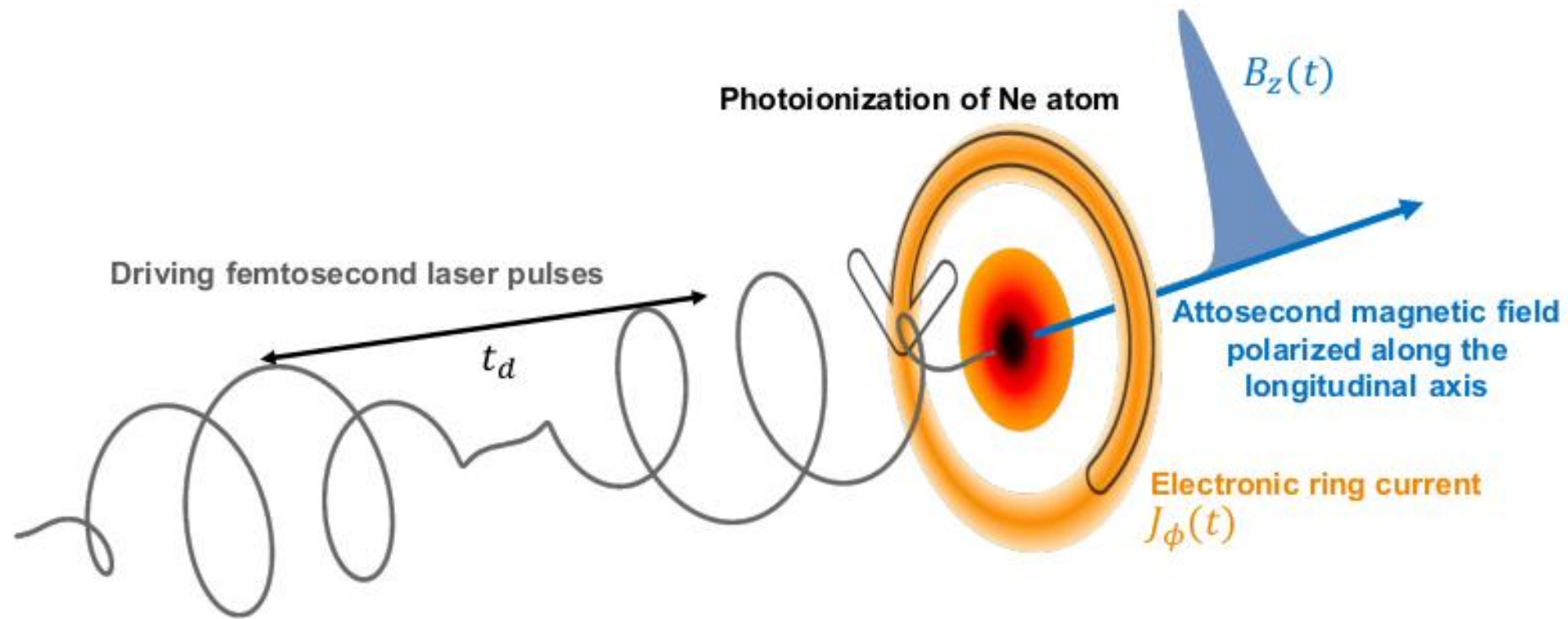
Ampère-Maxwell law

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

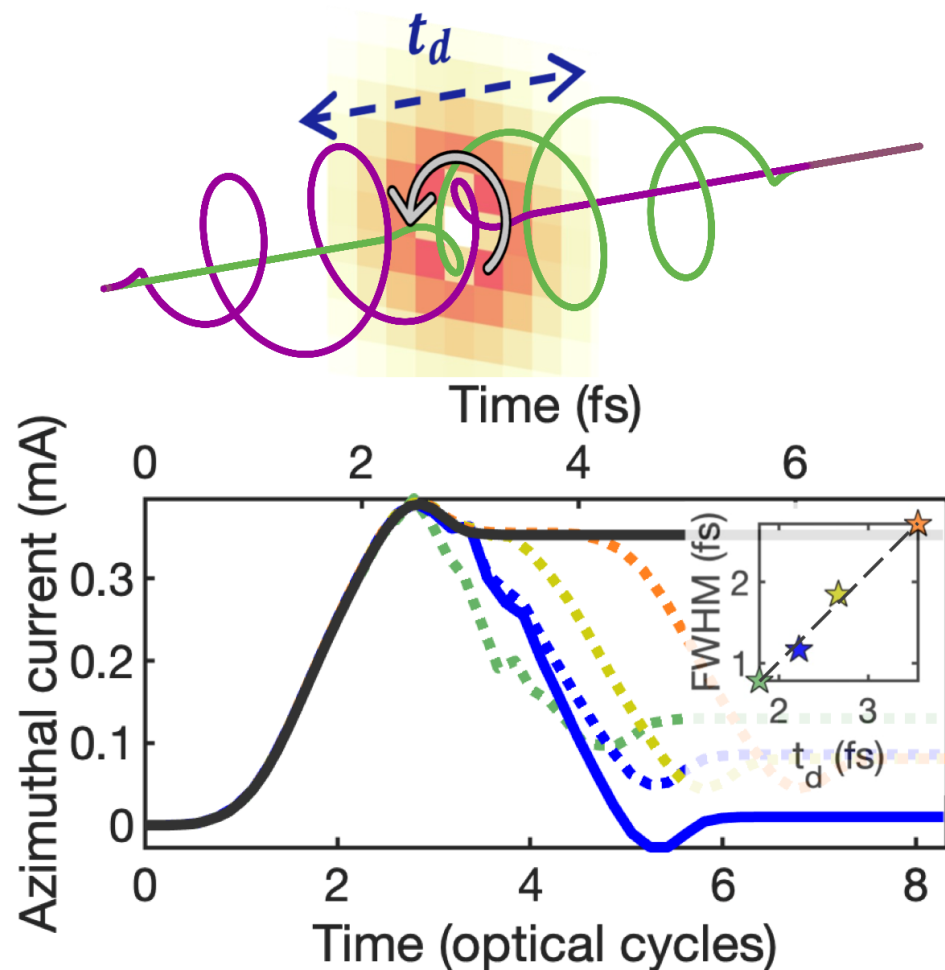


Generation of attosecond magnetic fields through ring-current gating



Tunable attosecond magnetic field waveforms via ring-current gating

Two gating femtosecond laser pulses

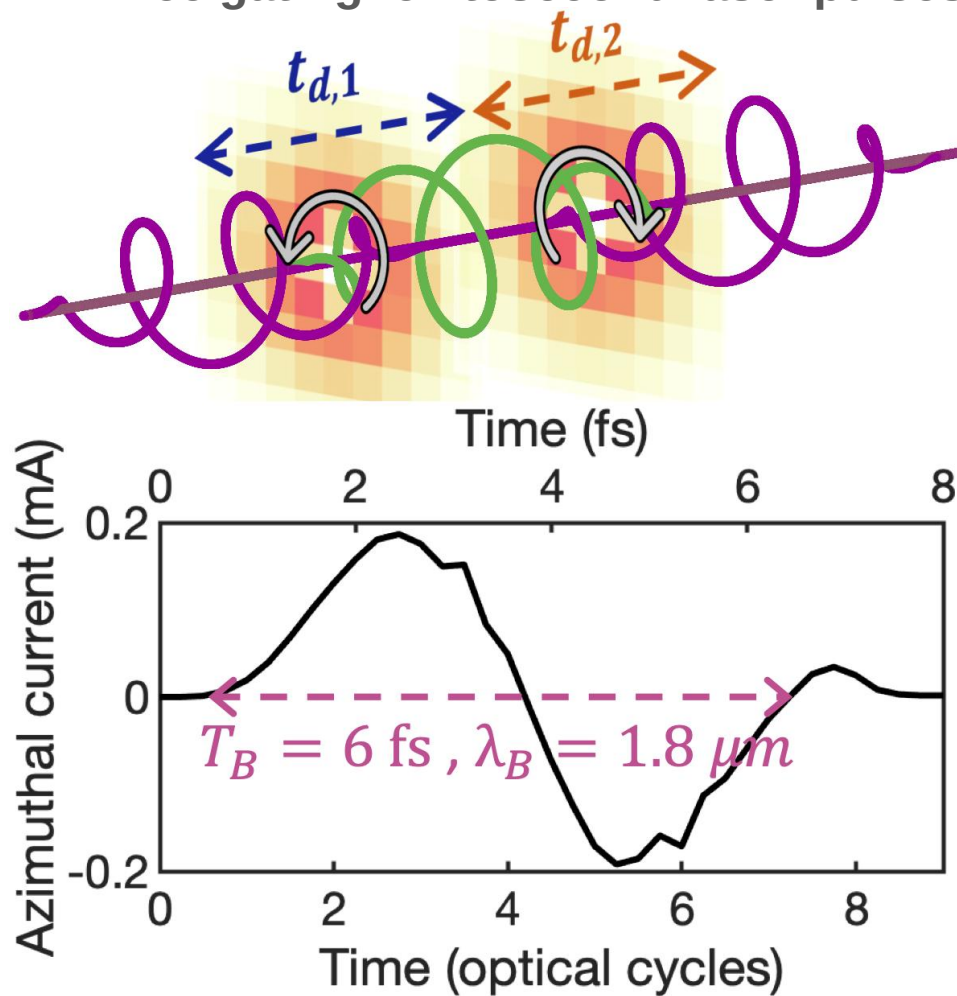


Options to isolate the magnetic field:

- Temporal isolation at time delays longer than the duration of the driving pulses
- Polarization filtering
- Low-frequency filtering

Tunable attosecond magnetic field waveforms via ring-current gating

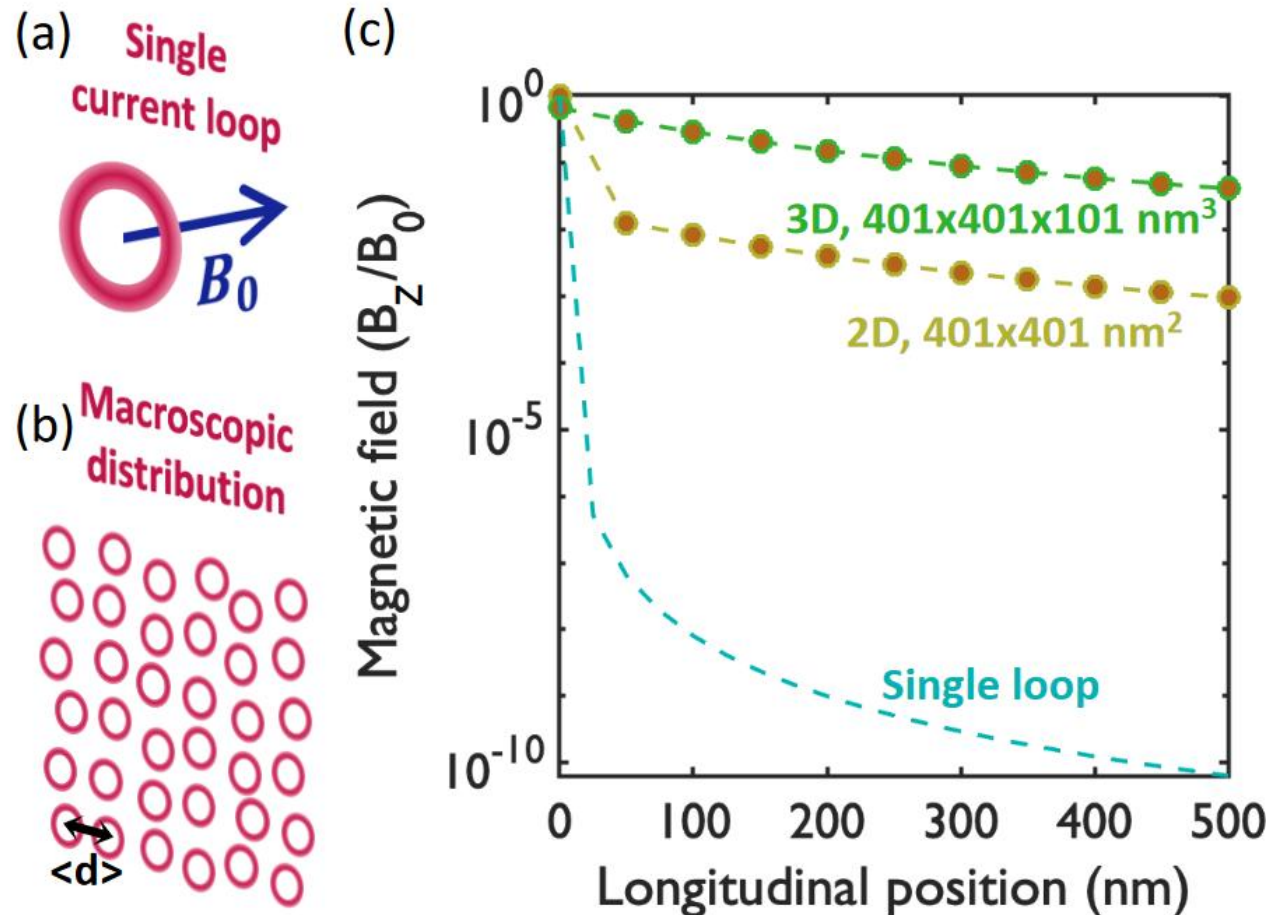
Three gating femtosecond laser pulses



- Synthesis of a single-cycle magnetic field with three gating pulse drivers
- Wavelength tunability by adjusting the time delay between the gating pulses

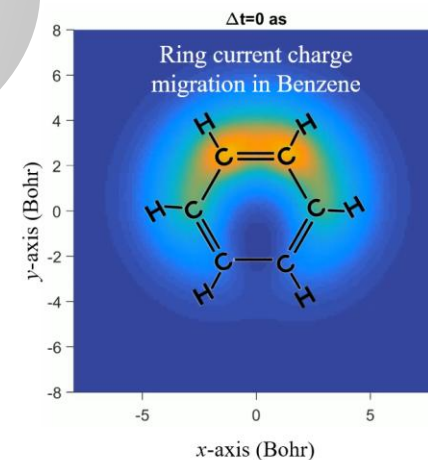
Macroscopic scaling of the generated magnetic field

What happens in the 3D gas jet? (in reality we don't have a single atom)

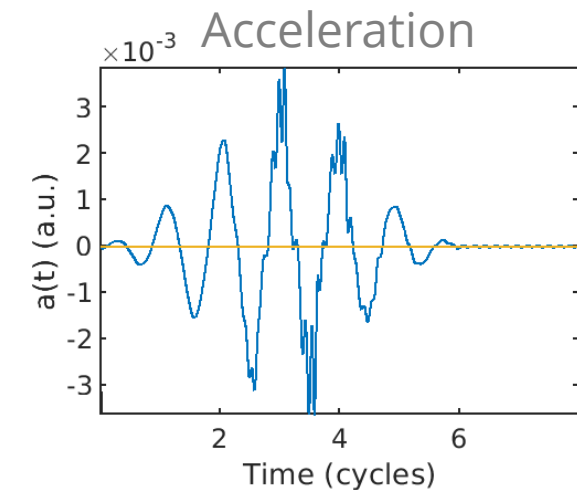
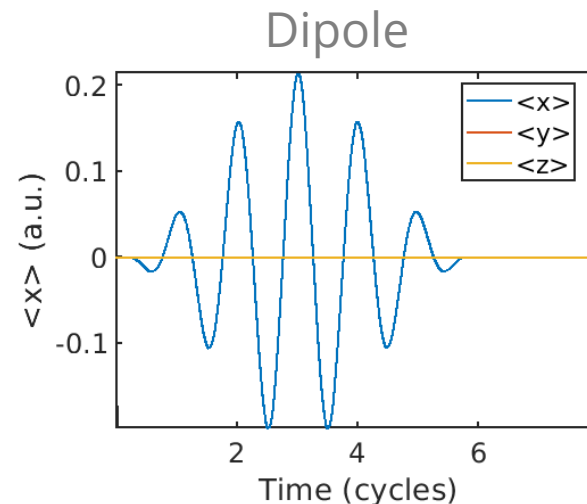
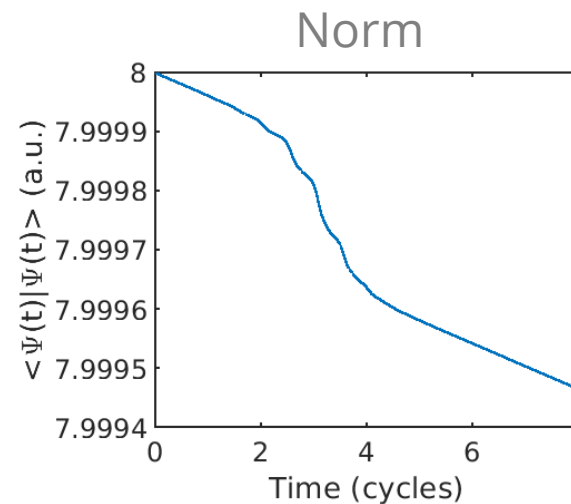
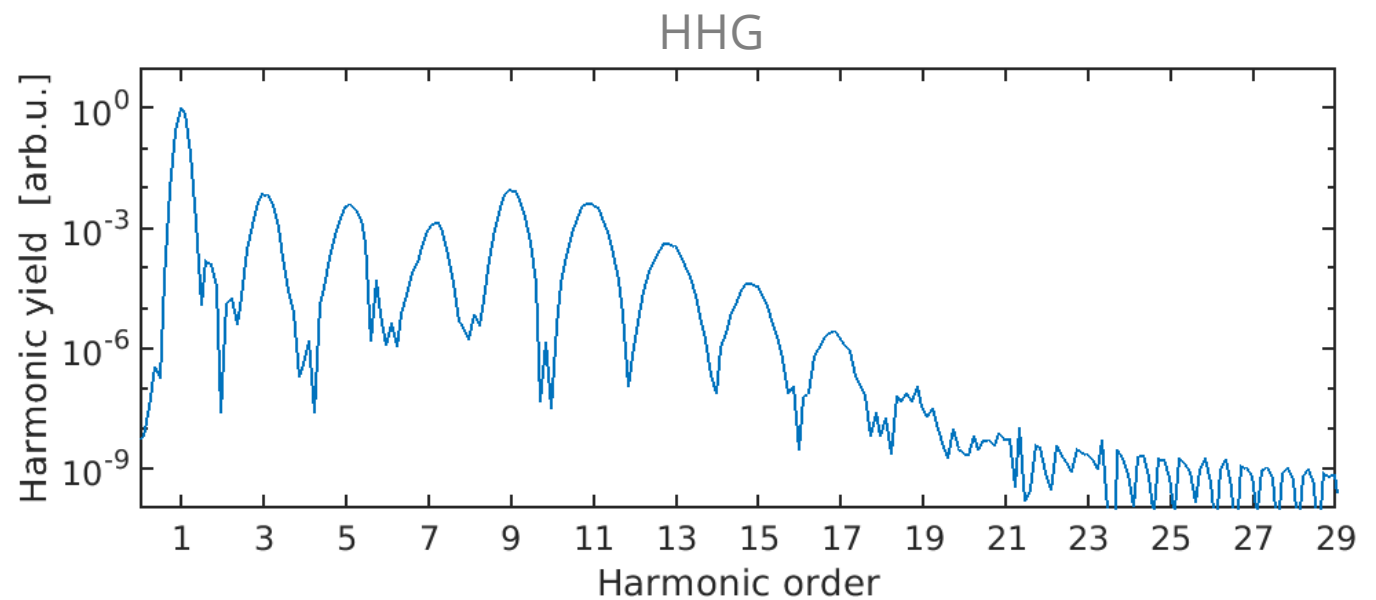
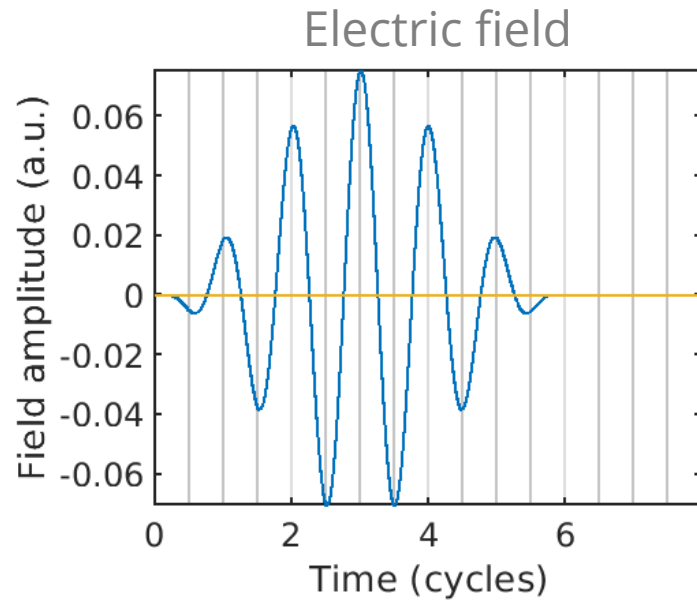


- Single-loop result follows the Biot-Savart law
$$B_z \propto 1/(z^2 + a^2)^{3/2}$$
- The 3D-gas medium presents a smooth decay along the longitudinal position.
- The magnetic flux at 100 nm from the gas jet can be as high as 1 T.

- **First scheme for generating attosecond magnetic fields**
- **High control of the temporal waveforms**
- **Tesla-scale magnetic flux at a hundred nanometers from the source**
- **Potentially separable from driving electric/magnetic fields**
- **Future applications in ultrafast magnetic dynamics and chiral responses**
- **Similar schemes can be implemented in molecules and solids for novel spectroscopies.**



E2: HHG in Ne atom driven by a LP driving pulse



Autoevaluation form of SESSION 2

<https://forms.gle/RM27UFg4PKB9zxwN6>

